

Q1 - 26 June - Shift 2

A batsman hits back a ball of mass 0.4 kg straight in the direction of the bowler without changing its initial speed of 15 ms^{-1} . The impulse imparted to the ball is _____ Ns.

Space for your notes:

Q2 - 27 June - Shift 1

Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) cm and (x, 0) cm respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is :

Space for your notes:

- (A) 4 cm towards the 10 kg block
- (B) 2 cm away from the 10 kg block
- (C) 2 cm towards the 10 kg block
- (D) 4 cm away from the 10 kg block

Q3 - 27 June - Shift 1

What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass? (Assume the collision to be head-on elastic collision)

Space for your notes:

- (A) 50.0% (B) 66.6%
- (C) 55.5% (D) 33.3%

Q4 - 27 June - Shift 1

Questions

MathonGo

A pendulum of length 2 m consists of a wooden bob of mass 50 g. A bullet of mass 75 g is fired towards the stationary bob with a speed v . The bullet emerges out of the bob with a speed $\frac{v}{3}$ and the bob just completes the vertical circle. The value of v is _____ ms^{-1} . (if $g = 10 \text{ m/s}^2$)

Space for your notes:

Q5 - 28 June - Shift 1

The position vector of 1 kg object is $\vec{r} = (3\hat{i} - \hat{j})\text{m}$ and its velocity $\vec{v} = (3\hat{j} + \hat{k})\text{ms}^{-1}$. The magnitude of its angular momentum is $\sqrt{x} \text{ Nm}$ where x is _____.

Space for your notes:

Q6 - 29 June - Shift 1

A body of mass M at rest explodes into three pieces, in the ratio of masses 1 : 1 : 2. Two smaller pieces fly off perpendicular to each other with velocities of 30 ms^{-1} and 40 ms^{-1} respectively. The velocity of the third piece will be :
(A) 15 ms^{-1} (B) 25 ms^{-1}
(C) 35 ms^{-1} (D) 50 ms^{-1}

Space for your notes:

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Answer Key

Q1 (12)

Q2 (C)

Q3 (C)

Q4 (10)

Q5 (91)

Q6 (B)

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Q1 (12)

Impulse = change in momentum

$$= m[v - (-v)] = 2mv$$

$$= 2 \times 0.4 \times 15 = 12 \text{ Ns}$$

Q2 (C)

$$\Delta x_G = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2}$$

$$0 = \frac{10 \times 6 + 30(\Delta x_2)}{40}$$

$$\Delta x_2 = -2 \text{ cm}$$

Block of mass 30 kg will move towards 10 kg.

Q3 (C)

Velocity after collision

$$V_2 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{m_1 + m_2}$$

$$V_2 = \frac{(5m - m)0 + 2m \cdot u_0}{m + 5m} = \frac{u_0}{3}$$

$$\% \Delta KE = \frac{\frac{1}{2} 5m \left(\frac{u_0}{3} \right)^2 - 0}{\frac{1}{2} m u_0^2} \times 100$$

$$= \frac{5u_0^2}{9u_0^2} \times 100 = \frac{500}{9} = 55.6\%$$

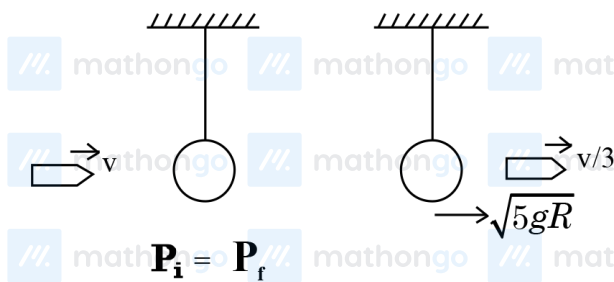
Q4 (10)

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Hints and Solutions

MathonGo

Considering Only Horizontal direction



$$\mathbf{P}_i = \mathbf{P}_f$$

$$(75v) + 0 = 50(\sqrt{5gR}) + 75\frac{v}{3}$$

$$75\left(v - \frac{v}{3}\right) = 50\sqrt{100}$$

$$v = 10 \text{ m/s}$$

Q5 (91)Using $\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$, $m = 1\text{kg}$

$$\vec{L} = (3\hat{i} - \hat{j}) \times (3\hat{j} + \hat{k}) = (9\hat{k} - 3\hat{j} - \hat{i}) \text{ N-s}$$

$$\Rightarrow |\vec{L}| = \sqrt{91} \text{ N-s}$$

Q6 (B)

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Mass of pieces by $\frac{M}{4}, \frac{M}{4}, \frac{M}{2}$

conserving momentum

$$\vec{P}_1 + \vec{P}_2 + \vec{P}_3 = 0$$

$$\vec{P}_3 = -(\vec{P}_1 + \vec{P}_2)$$

As $\vec{P}_1$ & $\vec{P}_2$ are perpendicular

$$\text{so } P_3 = \sqrt{P_1^2 + P_2^2}$$

$$P_3 = (50) \frac{M}{4}$$

$$\& P_3 = \frac{M}{2} v$$

$$\text{so } v = 25$$